Project Proposal
RehabSense – Smart Wearable Knee Rehabilitation Tracker

01. Background & Motivation

Musculoskeletal injuries and post-surgical rehabilitation cases such as Anterior cruciate ligament reconstruction repair, knee replacement require consistent physiotherapy exercises. But, most of the times patients perform exercises incorrectly at home, clinicians do not have enough objective data on actual range of motion, progress tracking is subjective and inconsistent.

Traditional method for measuring joint or knee angle is manual use of goniometer. But it can't measure the angle accurately and continuously while moving. Physiotherapists are unable to track improvement of the patient's movements, without continuous measurements. Also, no data is collected when patients perform exercises at home.

There is a need for an affordable, wearable system that providing objective, real-time joint angle measurement and repetition tracking to support home based rehabilitation.

02. Project Objective

The objective of RehabSense is to design and implement a wearable, rechargeable, embedded rehabilitation monitoring prototype that can,

- Measure knee joint angle using two IMU sensors
- Calculate range of motion
- Detect and count physiotherapy repetitions
- Stream data wirelessly to a dashboard
- Visualize recovery progress

03. System Overview

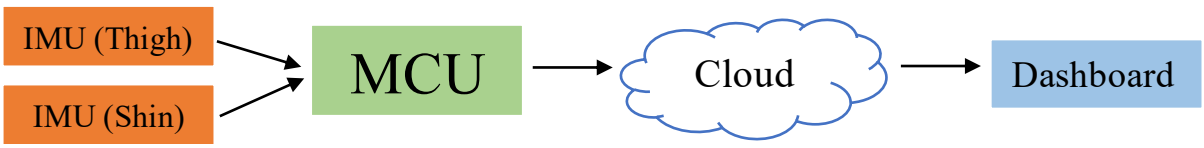

Figure 1. Simple Input ►Output flow

Knee angle can be estimate by using,

Knee angle ≈ Thigh Pitch – Shin Pitch

The system is using calibrated IMU tilt estimation to compute joint angle. Machine learning methods can be applied to classify exercise quality, detect incorrect movements, and analyze rehabilitation progress.

04. Hardware Design

There are two separate units in this design. One is the main unit which is attaching to thigh strap and the other is shin unit which is attaching to shin strap. There are separate components that are integrating in each unit.

Main Unit (Thigh Strap)

- ESP32 microcontroller
- Rechargeable Li-ion battery (2500-3000mAh)
- TP4056 charging module (with protection)
- MT3608 boost converter
- Power switch
- MPU6050
- Custom enclosure (3D printed)

Secondary Sensor (Shin Strap)

- MPU6050 IMU
- I2C wired connection to main unit
- Compact enclosure

All the components will be fully soldered and the prototype will be clean and suitable for demonstrations.

05. Relevance and sustainability

This project is relevant to,

- Remote rehabilitation
- Digitalized physiotherapy
- Preventive healthcare technology

This project is giving a sustainable solution in healthcare sector. This project promotes sustainability by,

- Supporting home based rehabilitation and reducing clinic visits
- Reducing transportation related emissions
- Using low cost reusable electronics
- Tracking recovery to reduce long-term healthcare burden

06. Required equipment

Electronics

- ESP32 board
- 2 × MPU6050
- Rechargeable Li-ion battery
- TP4056 charger module (with protection)
- Slide switch
- Dotboard
- Thin silicone wires
- JST connectors
- Heat shrink tube

Mechanical

- 3D printed enclosures
- Snap hook elastic binding strap
- Rubber padding
- Small screws

Tools

- Soldering iron
- Multimeter
- Hot glue gun